

Column	Row	D24HourMean	QuarterlyMean
1	29	11.95	11.95
1	31	10.69	10.69
1	37	11.66	11.66
1	39	10.22	10.22
1	41	10.91	10.91
1	45	10.90	10.90
1	49	12.01	12.01
1	53	10.12	10.12
1	57	11.12	11.12

3.3.2 Maps

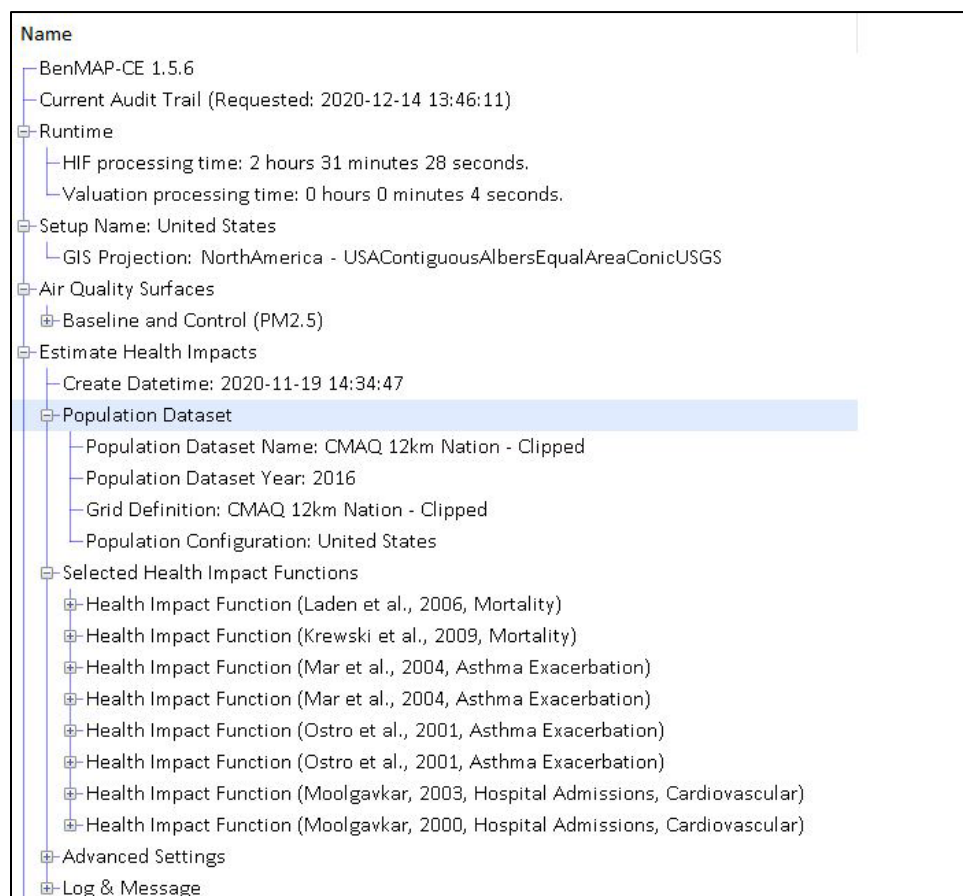
GIS Maps can be viewed in the lower right frame of the BenMAP-CE main window. Once an air quality surface is displayed, you can choose which layers to view by selecting or deselecting items in the GIS table of contents. Spatial layers (except regional administrative layers) are semi-transparent so that overlapping layers are viewable. You can export a formatted graphics file (e.g., .png format) with the map legend and title. For detailed instructions on GIS maps, see Chapter 8: GIS/Mapping.

3.3.3 Audit Trail Report

The **Audit Trail Report** provides a summary of the options selected in the various parts of the analysis. You may generate an audit trail with any of the file types used in BenMAP-CE: Air Quality Grids (with the .*aqgx* extension), Configurations (with the .*cfgx* extension), Configuration Results (with the .*cfgrx* extension), Aggregation, Pooling, and Valuation Configurations (with the .*apvx* extension), and Aggregation, Pooling, and Valuation Configuration Results (with the .*apvr*x extension). The report itself has a tree structure. Below is an example of an Audit Trail Report in the default output format.

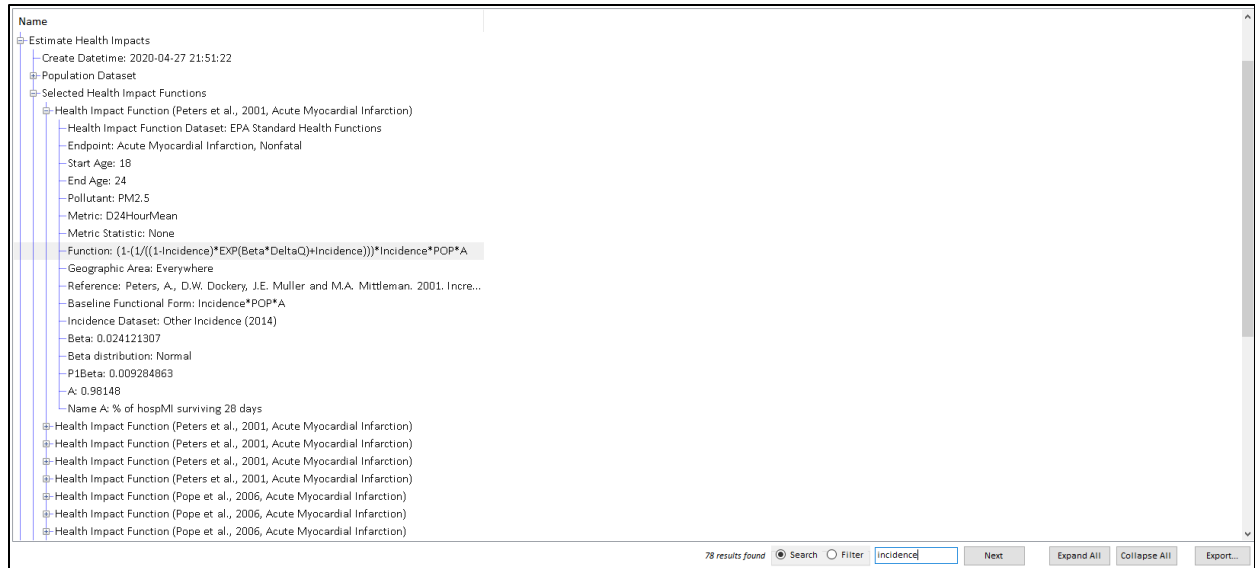
Name
BenMAP-CE 1.5.7
Current Audit Trail (Requested: 2020-12-14 13:43:07)
⊕ Runtime
⊕ Setup Name: United States
⊕ Air Quality Surfaces
⊕ Estimate Health Impacts
⊕ Aggregate, Pool & Value

Note that each successive step in an analysis contains a summary of its inputs and attributes, and those of each previous step in the analysis, laid out to follow the sequence of the inputs provided by the user. Below is an example of an **Audit Trail Report** when nodes have been expanded.



For example, in the above report the attributes of the *Health Impact Function* file used to generate the APV Results are present in the *Estimate Health Impacts* then *Selected Health Impact Functions* node. Similarly, the metadata for both the baseline and control air quality grids are present under the *Air Quality Surfaces* node.

The **Audit Trail Report** is now searchable. You may type a string into the Search text box at the bottom right of the **Audit Trail Report** and the program will highlight the location of that string within the report. Below is an example of the Search function.



For more information on audit trails, see Chapter 8: GIS/Mapping.

3.3.4 Dataset Validation Reports

You may load data to BenMAP-CE to tailor the analysis to your specific needs (click **Modify Datasets** from the main menu). Loading data requires specific formatting. BenMAP-CE offers a validation option to confirm that the proper headings and data types are present in the selected file. The validation routines also check that values are within reasonable ranges for certain types of data. If the file does not meet the validation requirements, error and/or warning messages will be reported. For more information about loading and validating data, see Chapter 4: Loading Data.

3.3.5 File Types

BenMAP-CE has a number of file types that you can use to store the settings used in a BenMAP-CE analysis, the results of an analysis, as well as maps and reports. Table 3-1 presents the names of the different file types, their functions, and their default folder locations.

Table 3-1. File Types Generated by BenMAP-CE

File Extension	Description	Default Folder Location¹
*.aggx	Air quality grid.	Result\AQG
*.apvx	Aggregation, Pooling, and Valuation configuration specifying the aggregation levels, pooling options, and valuation methods used to generate aggregated incidence estimates, pooled incidence estimates, valuation estimates, aggregated valuation estimates, and pooled valuation estimates.	Result\APV
*.apvrx	Aggregation, Pooling, and Valuation configuration results, containing incidence results at the grid cell level, aggregated incidence results, valuation results, aggregated valuation results, and pooled valuation results.	Result\APVRX
*.bdbx	BenMAP-CE Database Export tool creates files which can contain individual datasets or entire setups. These are saved in a specific format for importing to BenMAP-CE.	User-specified
*.cfgx	Configuration specifying the health impact functions and other options used to generate incidence estimates.	Result\CFG
*.cfgrx	Configuration results, containing incidence results at the grid cell level.	Result\CFGR
*.csv	Reports (Results Tables) and PopSim tool results are exported as *.csv files, which may be viewed in a text editor, or in programs such as Excel. Users may also export portions of BenMAP setups (e.g., baseline incidence rates) as .csv files.	Result\CFGRX or Result\APVRX, PopSim
*.projx	Project files save a BenMAP configuration (not results) so that users can reload analysis parameters without having to repeat the analysis steps. The configurations for the Health Incidence and Aggregation, Pooling, and Valuation steps may be saved simultaneously in a single project file.	Result\Project
*.rtf	Validation results from data imports	ValidationResults
*.shp	Shape files generated by BenMAP-CE's geographic information maps system. These files can be viewed within BenMAP-CE or within shape file viewers, such as ArcView.	AppData\...\Shapefiles ²
*.xlsx	GBD Rollback Tool results are exported as .xlsx files, which may be viewed in a spreadsheet tool such as Excel (there is also a .csv option).	GBD

¹ Most files generated by BenMAP-CE are stored within the User's directory under C:\Users\<user name>\Documents\My BenMAP-CE Files\.

² Shape files (*.shp) are stored at C:\Users\<user name>\AppData\Local\BenMAP-CE\Data\Shapefiles\<setup name>\.